Analyzing the Impact of Macroeconomic Surprises on S&P 500 Daily Returns

DS 4420 Final Project

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I. Introduction and Literature

Macroeconomic indicators play a central role in shaping financial markets by providing timely information on the health and direction of an economy. In the United States, indicators such as inflation releases, non-farm payrolls (NFP), and retail sales are closely monitored by economists, policymakers, investment banks, and portfolio managers. Prior to each release, professional forecasters submit their expectations to Bloomberg, which publishes a consensus forecasts representing the aggregated views of major financial institutions and research groups. Market participants often treat these published expectations as "baseline" values, and asset prices frequently respond not to the actual level of the indicator itself, but to the surprise or the difference between the realized value and the forecast. Unexpectedly strong job growth or decline, an unanticipated acceleration in inflation, or sudden weakness in consumer spending can move U.S. equity indexes, Treasury yields, and other financial assets within seconds of release. Despite this widespread market belief that macroeconomic surprises "move markets," the actual magnitude, and statistical strength of this relationship (particularly for daily S&P 500 returns) remains a more of an open question. Our project aims to quantify this impact.

A substantial amount of academic literature study the relationship between macroeconomic conditions and U.S. equity performance. One class of studies focuses on the effect of changes in macro variables rather than expectations-based surprises. For example, *Assessing the Impact of Macroeconomic Variables on the Performance of the U.S. Stock Market* uses multiple linear regression with quarterly data from 2010-2020 and reports that inflation, GDP growth, and unemployment rates show statistically significant relationships with index performance. Another study, *Macroeconomic Predictors and Stock Market Dynamics of the U.S. Equity Market*, uses time-series models such as GARCH, Granger causality tests, and Johansen cointegration over data from 2000-2023. It finds that GDP growth and interest rates are the most reliable predictors of broad equity returns, while inflation and unemployment exhibit predictive power mainly when explaining volatility. These studies, however, do not incorporate forecast information and therefore do not isolate the market-relevant "surprise" component.

A second set of papers focuses explicitly on macroeconomic surprises, which as previously mentioned, is defined as Actual - Forecast. *Do U.S. Macroeconomic Surprises Influence Equity Returns?* uses FP-GARCH models to study U.S. and international markets from 1993-2011 and finds that surprises in unemployment, consumer sentiment, and industrial production significantly influence both return levels and conditional volatility. Notably, these effects are state-dependent, being larger during bear markets. A related study, *Market Impact of Macroeconomic Announcements: Do Surprises Matter?*, compares different surprise metrics, including Actual-Expected and alternatives (such as Actual(t) – Actual(t-1) and Expected(t) – Actual(t-1)) using ARMA-GARCH models for 2004-2014. Their results highlight that NFP, ISM manufacturing, and consumer credit surprises are consistently significant predictors of S&P 500 movements.

While these studies provide valuable insight, they share several common characteristics that leave meaningful research gaps. First, most rely on classical linear regression or ARMA-GARCH models, which offer point estimates but only limited ability to quantify uncertainty in how strongly macro indicators affect returns. None of the reviewed papers incorporate a Bayesian approach, which provides full posterior distributions for effect sizes and probabilities and is particularly well-suited for evaluating the likelihood and uncertainty of market reactions.

Our project contributes to the literature in two main ways. First, we compare a Bayesian linear regression model with a frequentist ARIMAX model to evaluate the impact of surprises in three major U.S. macroeconomic indicators (NFP, core PCE, and core retail sales) on daily S&P 500 returns. All these indicators represent different key parts of the US economy, with NFP representing the labor market, core PCE representing inflation, and core retail sales representing consumer spending. The Bayesian model provides posterior distributions for effect sizes, credible intervals, and scenario-based predictive distributions, enabling probabilistic statements about the likelihood that a macro surprise moves equity returns in a particular direction. The ARIMAX model complements this by incorporating autoregressive structure and allowing scenario analysis while remaining consistent with traditional econometric approaches used in prior research. Second, by examining the full reaction curves and predictive uncertainty surrounding each indicator, we develop a more transparent quantification of how sensitive S&P 500 returns are to macroeconomic surprises. Overall, we find both provide novel insight into the magnitude, sign, and uncertainty of U.S. equity market reactions to unexpected macroeconomic information.

II. ML Methodology and Data Collection

To analyze how macroeconomic surprises influence daily S&P 500 returns, we assembled a dataset combining historical U.S. macroeconomic indicators, their associated Bloomberg forecast values, and equity market returns. Historical macroeconomic data, including both the realized releases and Bloomberg consensus forecasts, were obtained directly from the Bloomberg Terminal. We exported monthly time-series data for each indicator from December 2009 through the most recent available observations as of our project start date (September 26, 2025).

We focused on three indicators that represent the core pillars of U.S. macroeconomic health: labor markets, inflation, and consumer spending. Non-Farm Payrolls (NFP) is widely regarded as the most influential labor statistic, measuring the monthly change in U.S. employment while excluding agriculture, private household workers, and volunteers. Its releases frequently generate substantial market reactions due to its relevance for economic growth and Federal Reserve policy expectations. Core Personal Consumption Expenditures (Core PCE) was selected as our inflation indicator because it excludes volatile food and energy components and is the Federal Reserve's preferred gauge of inflation. Using the core version allows us to better isolate the "true signal" in inflation developments without excess noise from seasonal or commodity-driven fluctuations. Core retail sales, which similarly excludes volatile categories such as gas, autos, and food services, was chosen to capture consumer spending trends. Though GDP is often used in academic work, its quarterly frequency produces far fewer "surprise" events, making it less suitable for daily return analysis within our modeling framework.

For equity market data, we collected daily S&P 500 log returns from the Federal Reserve Bank of St. Louis (FRED), a widely used and highly reputable source for financial time-series datasets. The S&P 500 index represents the 500 largest publicly traded U.S. companies and is one of the most commonly used benchmarks of the U.S stock market. Although several papers in our literature review also examined the S&P 500, our project differentiates itself by evaluating both a frequentist time-series approach (ARIMAX) and a Bayesian linear regression model, addressing the same underlying economic question through two distinct lenses.

One of the methods covered in our project was the ARIMAX model, this model is an extension of the ARIMA model, which explains a time series using its own past values and past forecast errors. The ARIMA(p, d, q) structure models the differenced or stationary series yt through autoregressive terms, moving-average terms, and differencing operations. Formally, an ARIMA(p, d, q) process can be expressed as

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

To incorporate macroeconomic surprises directly into the model, we extend ARIMA to ARIMAX, where the "X" denotes the inclusion of exogenous regressors. In our case, these regressors are the standardized macroeconomic surprises, NFP, core CPI, and core retail sales, observed on the days they are released. Therefore, the ARIMAX model adds on to the classical ARIMA structure by adding a regression component that captures the effect of these external variables on daily S&P 500 returns. The full ARIMAX equation therefore becomes

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \beta^T x_t + \epsilon_t$$

where x_t represents the vector of macro surprises at time t, and β measures their marginal impact on returns. This structure allows the model to jointly capture both the autoregressive and moving average behavior of equity returns and the exogenous influence of unexpected macroeconomic reads. In other words, the adjustment from ARIMA to ARIMAX enable the model to quantify how shocks to inflation, employment, or consumer spending translate into movements in the S&P 500.

The second method employed is Bayesian linear regression, where the likelihood is the distribution of S&P return given the macro-indicator surprises, in which the likelihood is assumed to follow normal distribution. We first created the training and testing subset with a test size of 0.25, then used them to define a prior distribution that models the weights of the linear regression best fit line. Each weight corresponds to the coefficient for each indicator and the intercept. Because the conjugate prior for a Gaussian likelihood is also a Gaussian distribution, normal distribution is used as the prior distribution for our model. The prior distribution is in 4th dimension because the total number of features is 4 (1 intercept + 3 indicators), and thus the length of the prior mean vector is 4 and the shape of the covariance matrix is 4 × 4. We set the prior mean of the 3 indicator weights to zero, because historically each indicator only has minimal impacts on the daily S&P return, typically around $\pm$ 1%. Hence, our best initial belief is that these indicators' true impact on S&P return is around 0 before seeing concrete evidence from the data. Similarly, the prior mean of the intercept is set to zero as well, as the daily S&P return usually fluctuates slightly around zero unless significant market shocks occur.

For the covariance matrix, an identity matrix is used as we assume there's no correlation between the indicators themselves, hence the identity matrix is chosen as it sets the covariance between all indicators to be zero, while leaving the variance of each feature on the main diagonal that's initially set to one. The variance of each feature is then scaled with a constant of $\frac{1}{\alpha}$ where α is the prior precision hyperparameter, which controls how strongly the indicator weights are shrinking toward zero. Specifically, a larger α leads to a smaller variance that restricts the indicator's weight closer to its mean (zero), while a smaller α leads to a larger variance that allows the indicator's weight to be more varied and widespread around its mean. We set α to be 0.01 and thus the variance of indicators becomes 100. This small α is used because the impact of NFP, PCE, and retail sales on S&P return is not exactly at 0, but usually somewhere around it (ex: $\pm$0.01). Thus, a small α allows the indicator weights to vary more from zero to reflect the real-world variations. However, this constant of $\frac{1}{\alpha}$ isn't applied to the intercept variance. This is because we don't want to shrink the intercept too much as it serves as the baseline S&P return independent of all surprises, which is small but usually not exactly zero. By setting the intercept variance to infinity, we essentially make it uninformative and allow the data to determine the intercept freely.

The above computation leads to the below prior mean and covariance matrix for the prior distribution:

Prior mean: $[0,\ 0,\ 0,\ 0]$

Prior covariance: [[∞ 0, 0, 0]
[0, 100, 0, 0]
[0, 0, 100, 0]
[0, 0, 0, 100]]

Using the prior covariance, prior mean, data standard deviation, X_train, and y_train, the posterior distribution mean and covariance matrix is then defined as below:

$$\text{Posterior covariance} = \Sigma_N = \left(\Sigma_0^{-1} + \frac{1}{\sigma^2} X^T X\right)^{-1}$$

$$\text{Posterior mean} = \mu_N = \Sigma_N \left(\Sigma_0^{-1}\mu_0 + \frac{1}{\sigma^2} X^T y\right)$$

Specifically, the posterior mean and covariance are as below:

Posterior mean: [4.404e-04, 1.084e-03, 1.678e-05, 2.027e-04]

Posterior covariance: [[4.188e-08, 6.695e-09, -3.819e-10, -1.079e-09]
[6.695e-09, 5.205e-07, -3.401e-11, 9.847e-11]
[-3.819e-10, -3.401e-11, 4.27e-08, -5.24e-11]
[-1.079e-09, 9.847e-11, -5.24e-11, 5.254e-08]]

With the posterior mean and covariance being all set, the next step is to utilize them to investigate the impacts of macro-indicator surprises on the S&P return.

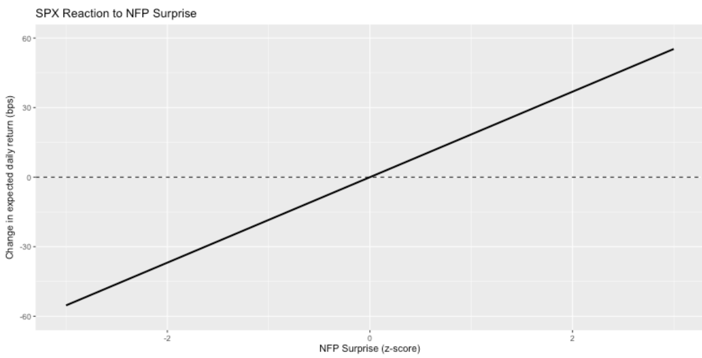
III. ML Results and Interpretation

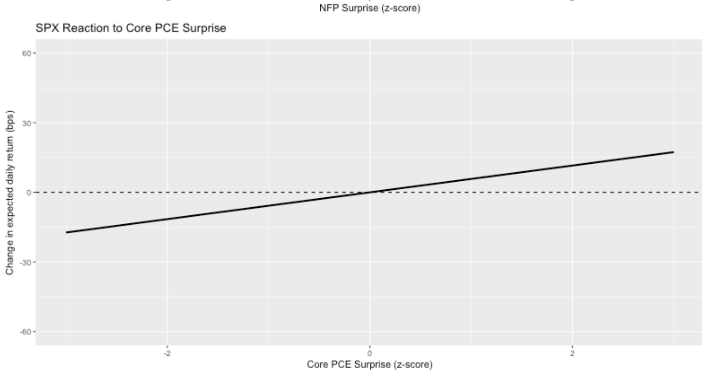
Before reporting the results of the ARIMAX analysis, we first validated the model on a training set (accounting for approximately 80% of the data). When analyzing the error metrics on the test set we found that the RMSE was ~0.0096, the MAE was ~ 0.0066, and the DA (Directional Accuracy), was ~0.5533. These results appeared reasonable based on our prior knowledge and allowed us to proceed forward with implementing the ARIMAX on the full data set. When examining the model's residuals, the residual series appears broadly stationary, though it contains several noticeable spikes. The ACF plot shows no statistically significant autocorrelation at short lags, indicating that most near-term relationships have been captured. However, several significant spikes appear around lags 22–26, suggesting that some medium-horizon dependence remains unexplained. This potentially reflects slower-moving market dynamics or volatility clustering. The residual histogram further shows a fat-tailed distribution relative to the normal density, which is consistent with the heavy tails typical of equity returns and suggests that a student-t distribution could better reflect the underlying data.

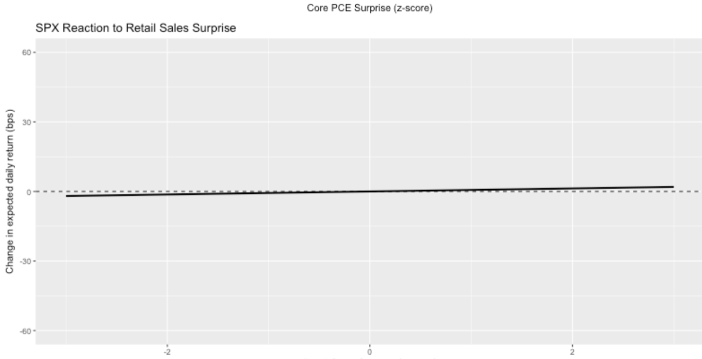
When interpreting the fitted ARIMAX(4,0,4) model, the dominant conclusion remains that daily S&P 500 returns are overwhelmingly driven by noise rather than macroeconomic fundamentals. However, the estimated coefficients indicate that macro surprises do have measurable, though still small, effects. A +1 standard deviation NFP surprise increases expected daily returns by roughly 18 bps**,** surprises in core PCE move returns by about 6 bps, and core retail sales surprises have a near-zero effect. Given that the model-implied daily volatility is approximately 1.07%**,** even these relatively large coefficients translate into only small shifts in the full predictive return distribution. Therefore, while the ARIMAX model detects statistically meaningful reactions, especially for NFP, the overall explanatory power of macroeconomic surprises for day-to-day equity index movements remains limited.

To illustrate these effects, we compared scenario plots for +2 and –2 standard-deviation surprises against a baseline day (where all indicators are set at their average surprise). The baseline expected return is 0.1219%, while a +2 SD NFP surprise increases the expected return to 0.4906%, and a –2 SD surprise lowers it to –0.2468%. For core PCE, the expected return under a +2 SD surprise is 0.2373%, whereas a –2 SD surprise produces only 0.00645%. The effects are smaller for core retail sales: the +2 SD scenario yields 0.1349%, and the –2 SD scenario yields 0.1088%, highlighting its minimal influence on daily equity returns.

Because ARIMAX only provides point forecasts and confidence intervals, we constructed full predictive distributions by backing out the implied standard deviation from the 95% forecast interval and sampling 10,000 draws from a student-t distribution with 5.5 degrees of freedom (estimated via MLE). A t-distribution was chosen instead of a normal distribution because equity return distributions are well-known to be heavy-tailed and therefore are better modeled using this type of probability distribution.







To further reinforce the relationship explained above, an interesting relationship that highlights the “importance” of each indicator regarding how they impact S&P 500 returns is shown to the left with these reaction curves.

This plot highlights how expected daily returns change (in basis points) as macroeconomic surprises range from –3 to +3 standard deviations. The steepest reaction curve belongs to NFP, where a 1 SD shock moves expected returns by roughly 18 bps, generating nearly ±55 bps of impact at the extremes ($z = \pm 3$). Core PCE produces a much smaller slope, with effects of about ±17 bps at $z = \pm 3$, while core retail sales exhibit an almost flat curve, contributing only about ±2 bps across the full range. These results closely mirror the coefficient estimates and reinforce the conclusion that, among the indicators examined, NFP surprises generate the largest movements in expected returns, though even these impacts remain quite small compared typical daily S&P 500 volatility.

Bayesian Model Results:

From section II, we know the posterior means for NFP, PCE, and retail sales. These values correspond to how much the S&P return is moved on average when each indicator is increased by 1 standard deviation. Below are the detail results:

On average, NFP surprise moves the S&P return by 0.00045
On average, PCE surprise moves the S&P return by 0.00006
On average, retail sales surprise moves the S&P return by 0.00005

This result means that on average, NFP surprise moves the S&P return the most as its effect size per 1 standard deviation is 0.045%, while PCE surprise has the effect size of 0.006% and retail sales surprise has the effect size of 0.005%. This indicates that NFP surprise is the most impactful to market return while the surprise of PCE and retail sales has less significant impact.

Next, the 95% credible interval of the indicator coefficients shows the range of coefficients for each indicator, where there's a 95% probability that the coefficient is within this range. Here's the credible interval for each indicator:

95% credible interval of the NFP surprise coefficient: (0.00010, 0.00080)
95% credible interval of the PCE surprise coefficient: (-0.00036, 0.00048)
95% credible interval of the retail sales surprise coefficient: (-0.00030, 0.00040)

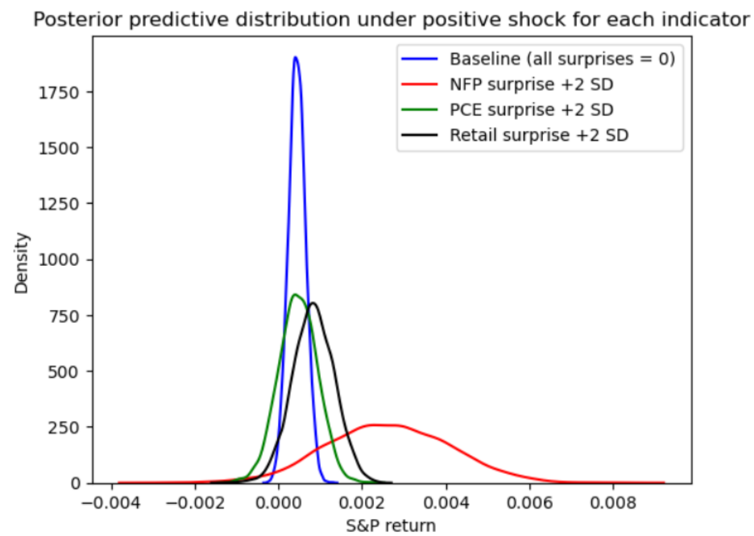
The credible interval of NFP surprise coefficient doesn't include zero as both its lower and upper bound are positive, meaning there's at least 95% probability that there's a positive impact on the S&P return. And since the coefficient is very unlikely to be zero, NFP is guaranteed to have impact on S&P return 95% of the time. The interval of PCE surprise coefficient includes zero as it ranges from a negative lower bound to a positive upper bound, meaning there's chance that PCE surprise has no impact on the S&P return at all when the coefficient is zero. The interval of retail sales surprise coefficient also includes zero with a negative lower bound and positive upper bound, meaning the coefficient could also potentially be zero and in turn not impacting the S&P return. This suggests that NFP is the only macro-indicator whose surprise has clear and significant impact on the S&P return, while PCE and retail sales do not show statistically meaningful effects as their credible interval includes zero.

In addition, we computed the probability that the coefficient for each indicator is positive, which indicates the probability that each indicator has a positive impact on the S&P return. Here's the probability results:

Probability that NFP surprise coefficient is positive: 0.9944
Probability that PCE surprise coefficient is positive: 0.6086
Probability that retail sales surprise coefficient is positive: 0.6078

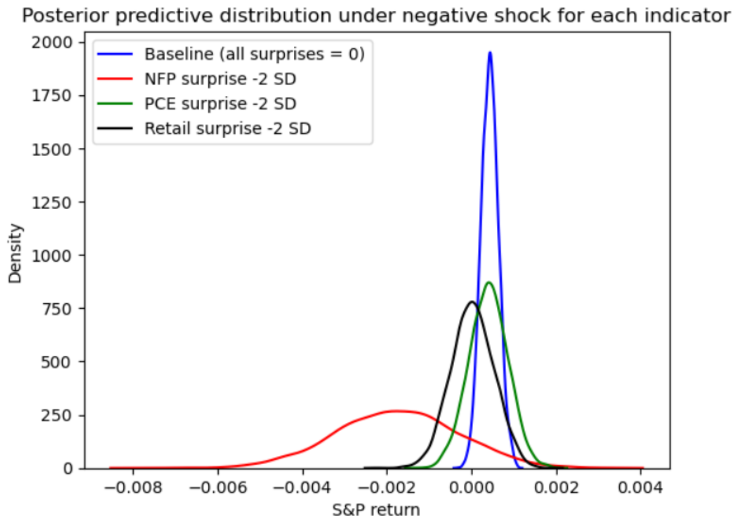
The probability that the coefficient of NFP surprise is positive is 0.9944, meaning there's a 99.44% probability that the NFP surprise has a positive impact on the S&P return, indicating a positive correlation between NFP surprise and S&P return. The probability that the coefficient of PCE surprise is positive is only 0.6086, meaning there's a 60.86% chance that PCE surprise has a positive impact on the S&P return, while 39.14% chance it has a negative impact. The probability that the coefficient of retail sales surprise is also only 0.6078, meaning there's a 60.78% chance that retail sales surprise has a positive impact on the S&P return, while 39.22% chance it has a negative impact. This suggests that the NFP surprise almost always has a positive impact on the S&P return, while PCE and retail sales surprise roughly positively impact the return 60% of the time and negatively impact the return 40% of the time.

Lastly, we did a scenario analysis on how the distribution of S&P return would look like when each individual indicator is increased or decreased by 2 standard deviations, while the rest of the indicators are holding constant. Here's the scenario analysis results:



Plot 1 shows the S&P return distribution under a positive shock for each indicator, where x-axis is the return and y-axis is density. The blue curve is the baseline return where all surprises are zero, which has a mean of 0.000439. It's narrow with most of its values centered around the mean. It's because when there's no surprise for all indicators, there's less uncertainty impacting the market, so S&P return is stable and doesn't fluctuate much.

The green curve corresponds to the S&P return when the PCE surprise is up by 2 standard deviations, which has the mean of 0.000467. This mean value is almost the same as the baseline mean, showing that PCE surprise has little impact on the S&P return as its increment doesn't shift the baseline mean much. The green curve is slightly wider than the baseline curve, indicating that a positive PCE shock still adds a little more uncertainty to the S&P return, making it vary more. The black curve corresponds to the S&P return when the retail sales surprise is up by 2 standard deviations, which has the mean of 0.000834. This mean value is higher than the baseline mean, showing that retail sales surprise has a positive impact on the S&P return by shifting the baseline mean to the right. Finally, the red curve corresponds to the S&P return when the NFP surprise is up by 2 standard deviations, which has the mean of 0.00262. This mean value is almost 6 times larger than the baseline mean, showing that NFP surprise has a strong positive impact on the S&P return by shifting the baseline mean to the far right. In addition, the red curve is also much wider than the baseline curve, indicating that NFP surprise adds a lot of uncertainty and variation to the market return, making the return very widespread.



The second plot has the exact same setup as the first plot, except it shows the distribution of the S&P return under a negative shock for each indicator. The blue curve again serves as the baseline S&P return with a mean of 0.000439, under the condition without any surprises. The green and black curves correspond to the S&P return when the PCE surprise and retail sales surprise decrease by 2 standard deviations. The green curve has a mean of 0.000408, very close to the baseline mean, meaning both increment and decrement in the PCE surprise has very little impact on the S&P return. The black curve has a mean of 0.0003, which is lower than the baseline mean, showing that retail sales surprise has a positive impact on the S&P return as the return goes down when the surprise decreases. Lastly, the red curve corresponds to the S&P return when the NFP surprise decreases by 2 standard deviations, which has the mean of –0.00174. This mean value is much smaller than the baseline mean and shifts the mean to the far left, showing that NFP surprise has a strong positive impact on the S&P return as the return goes down when the surprise decreases.

IV. Conclusions and Future Work

In conclusion, The ARIMAX analysis shows that while macroeconomic surprises do influence daily S&P 500 returns, their effects are small relative to the dominant noise in equity markets. A +1 SD surprise in NFP increases expected returns by roughly 18 bps, core PCE surprises move returns by about 6 bps**,** and retail sales surprises have near-zero impact. Even though NFP exhibits a statistically meaningful coefficient, the model-implied volatility of 1.07% per day means these shocks translate into only modest shifts in the overall predictive return distribution. Scenario analysis further confirms that even extreme ±2 SD surprises generate changes in expected returns that remain small relative to typical market fluctuations, reinforcing the conclusion that macro surprises don't have much explanatory power for short-horizon equity movements.

Looking ahead, several extensions could deepen and broaden these findings. First, future work could explore longer forecast horizons (such as 3-day, 1-week, or 10-day returns) where macroeconomic information may propagate more visibly. Second, extending the analysis to other asset classes (Treasuries, Forex, Commodities) or to other equity indices may reveal stronger macro sensitivities. Expanding the set of macro indicators or introducing interaction terms (for example, NFP surprises interacting with market regimes) could also capture nonlinear patterns not identified in the present model.

From an ARIMAX-specific perspective, the residual analysis suggests possibilities for more sophisticated modeling. Incorporating GARCH (volatility-modeling) would better account for the heavy tails and volatility clustering typical of financial returns. A regime-switching ARIMAX model, by categorizing periods of time as different "regimes", could capture structural breaks, such as differing macro sensitivity during COVID era versus non-COVID era. These enhancements would allow the model to offer an overall richer understanding of how macroeconomic surprises influence financial markets across different environments and perhaps lead to further interesting discoveries.

For the Bayesian model, the key findings are that NFP surprise has the largest effect size per 1 SD, while the effect size of surprise in PCE and retail sales are minimal, suggesting NFP surprise is the most impactful indicator shock to the S&P return. In addition, the 95% credible interval of NFP surprise doesn't include 0 and has a positive lower and upper bound, meaning we're 95% certain that NFP surprise will have a positive impact on the S&P return. For the PCE and retail sales surprise, both of their credible intervals contain zero and have a negative lower bound and positive upper bound. This means for PCE and retail sales surprise, their impact on the S&P return can be positive or negative depending on where the coefficient lands in the interval, or even have no impact on the S&P return at all as the coefficients can potentially be zero. Next, the probability that the coefficient of NFP surprise is positive is 0.994, so it's almost guaranteed that NFP surprise will have a positive impact on the S&P return. The probability for the coefficient of PCE and retail sales surprise to be positive is only 0.608 and 0.607, which is much lower and means these two indicator surprises can sometimes have positive impact while the other 40% of the time have negative impacts on the S&P return.

For future works, the first extension that can be implemented for the Bayesian model is to predict the S&P return by sampling coefficients from the posterior distribution. Once we have the coefficients, we can simply multiply it with the new indicator surprise data, which will give us the new predicted S&P return. This extension is nice as our project has only been analyzing the "impact" of the macro-indicator surprises on S&P return, but not "predicting" the S&P return using the indicators. Another extension is to analyze the correlation between the indicator surprises, in which finding a correlation means there exists dependency between the indicator surprises, hence the impact of these surprises can be redundant and overlapping. If the correlation is weak, then we can be sure that all indicator surprises are independent and don't affect each other. The last extension is to analyze the impact of indicator surprises separately instead of together, so we can isolate the impact of each individual indicator without the interference of other surprises. This can be crucial as we can analyze the impact of one single indicator surprise other than the combined impact of all indicator surprises, especially when there happens to exist dependency between the indicators.

V. Appendix: Implementation

Implementation of ARIMAX model (in R):

```r
# train-test split validation
cut_date <- as.Date("2022-12-31")

train_idx <- data$date <= cut_date
test_idx <- data$date >  cut_date

y_train <- y[train_idx]
y_test <- y[test_idx]
xreg_train <- xreg[train_idx, , drop = FALSE]
xreg_test <- xreg[test_idx, , drop = FALSE]

cat("Train size:", length(y_train), " Test size:", length(y_test), "\n")

# fit ARIMAX on training data only
fit_train <- auto.arima(y_train, xreg = xreg_train, seasonal = FALSE)

cat("Training model summary:\n")
print(summary(fit_train))

# out-of-sample forecast on test set
fc_test <- forecast(fit_train, xreg = xreg_test, h = nrow(xreg_test))

y_hat <- fc_test$mean

# basic error metrics
rmse <- sqrt(mean((y_hat - y_test)^2, na.rm = TRUE))
mae <- mean(abs(y_hat - y_test), na.rm = TRUE)
dir_acc <- mean(sign(y_hat) == sign(y_test), na.rm = TRUE)

cat("Out-of-sample performance;\n")
cat("RMSE:", rmse, "\n")
cat("MAE: ", mae,  "\n")
cat("Directional accuracy:", round(100 * dir_acc, 2), "%\n")

# refit on full data after satisfactory training performance
train_specs <- arimaorder(fit_train)
print(train_specs)
fit <- Arima(y, order = train_specs, xreg = xreg, seasonal = FALSE)

cat("\nFull-sample model:\n")
print(summary(fit))
```

Implementation of Bayesian linear regression (in python):

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25)

alpha = 0.01
feature_num = X_train.shape[1]

identity = np.eye(feature_num)
identity[0, 0] = 0
prior_cov = (1/alpha) * identity
prior_cov[0, 0] = np.inf
prior_mean = np.zeros(feature_num).reshape(-1, 1)

data_std = np.std(y_train)
prior_prec = np.eye(feature_num) * alpha
prior_prec[0, 0] = 0

cov_part_1 = np.matmul(X_train.T, X_train)
cov_part_2 = ((1/data_std**2)) * cov_part_1
cov_part_3 = prior_prec + cov_part_2
posterior_cov = np.linalg.inv(cov_part_3)

mean_part_1 = np.matmul(X_train.T, y_train).reshape(-1, 1)
mean_part_2 = (1/data_std**2) * mean_part_1
mean_part_3 = np.matmul(prior_prec, prior_mean)
mean_part_4 = mean_part_3 + mean_part_2
posterior_mean = np.matmul(posterior_cov, mean_part_4)

print (f"Prior distribution mean: \n{prior_mean}\n")
print (f"Prior distribution covariance matrix: \n{prior_cov}")
print (f"Posterior distribution mean: \n{posterior_mean}\n")
print (f"Posterior distribution covariance matrix: \n{posterior_cov}")
```

Citations:

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